



EDU TERIA

72nd BPSC

Prelims Online Batch

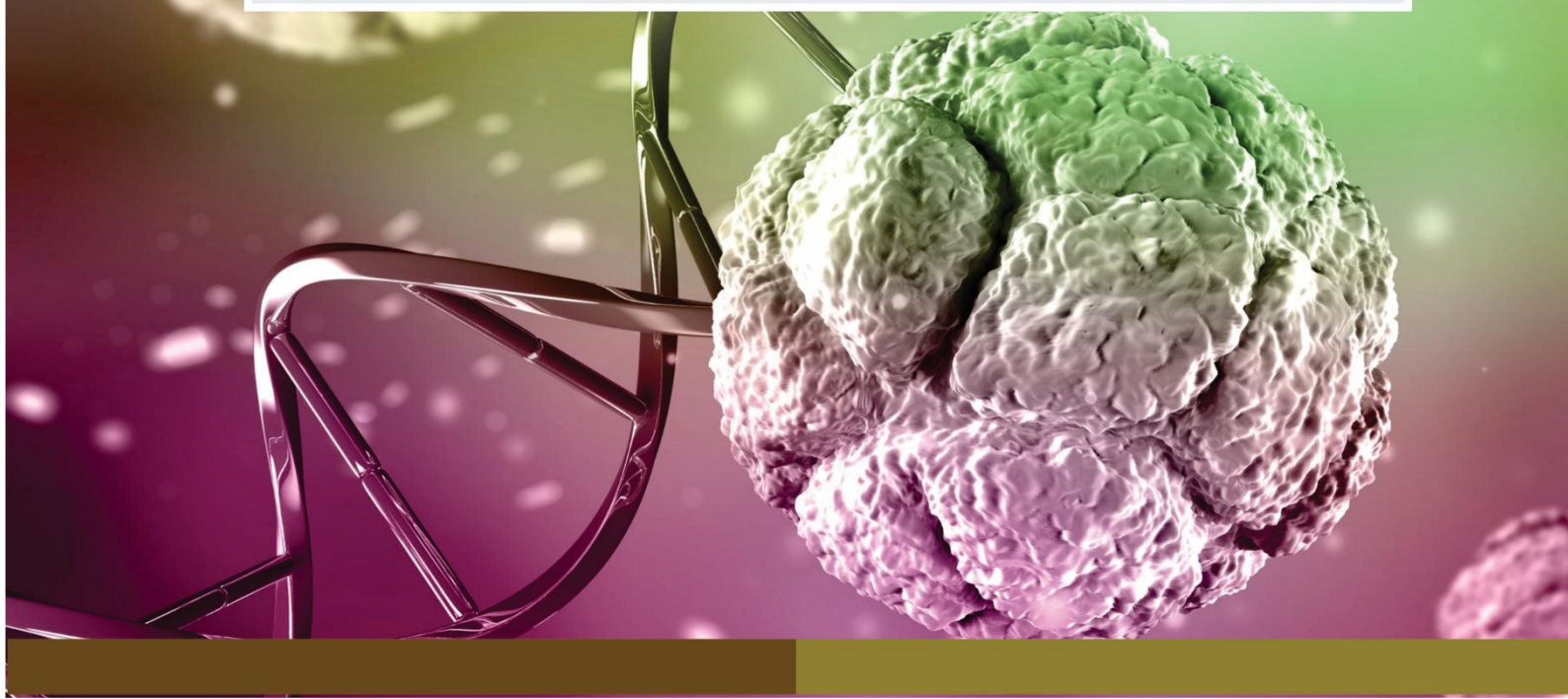
BIOLOGY

Topic wise
Notes

LECTURE

16

Circulatory System, Part - 02



Circulatory System, Part - 02

Body Fluids and Circulation

Blood:

Specialized fluid vascular connective tissue.

- It is a fluid connective tissue composed of different cells (RBCs, WBCs and platelets). The pH of blood is ~7.4, i.e. slightly alkaline.
- Study of blood → Hematology
- Formation of blood → Haemopoiesis
- Hormone stimulating haemopoiesis is known as Erythropoietin Hormone.

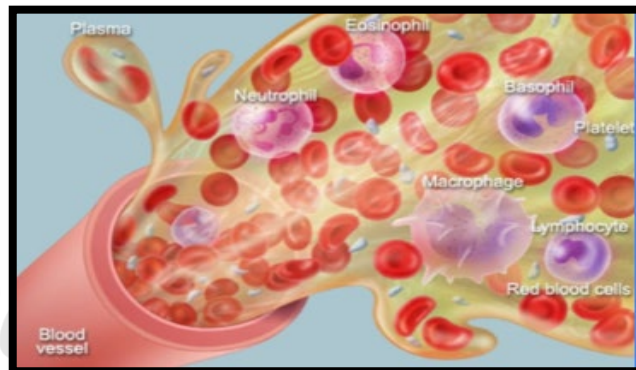
Average Blood Volume— 5-6 litre

Blood body Weight— 7-8 % of whole body.

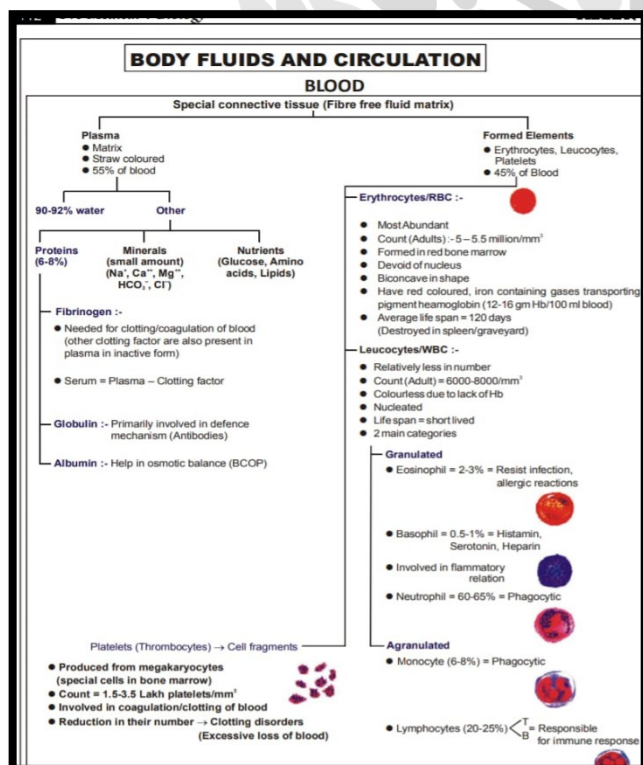
Blood is also called colloidal solution.

Blood forming organ— long bones and flat bones.

Human Blood



Characteristics	Acidophils	Basophils	Neutrophils	Monocytes	Lymphocytes
Size	10-14μ	8-10μ (smallest granulocytes)	10-12μ	12-20μ	6-16μ
Life span	14 hrs	10 hrs	12 hrs	less than 24 hrs in blood	5 to 7 days in blood
Stain with	acidic dye like eosin	Basic dye like methylene blue	any dye (acidic/basic/neutral)		
Shape of nucleus	Round	Two or three lobed, S-shape	3 to 5 lobed	Kidney / Bean shaped	Large due to which cytoplasm becomes peripheral
Function/s	They protect body against allergic and parasitic infections.	Secrete and transport heparin, histamine and serotonin	Phagocytic in nature. Destroy bacteria and viruses by phagocytosis.	Also called scavengers of blood because they engulf damaged or dead and minute bits of blood corpuscles.	- T-Killer - direct kill microbes - T-Helper - Stimulate - B-Lymphocytes to produce antibody - T-Suppressor - Suppresses - T-Killer and protect immune system. - B-Lymphocytes-Produce and transport antibodies.
Number	2-3% of TLC	0.5-1% (minimum in no.) of TLC	60-65% (maximum in no.) of TLC	6-8% of TLC	20-25% of TLC
Special point	Acidophils ↑ called Eosinophilia occurs during beritosis, Acantosis, Hay fever		- Due to their smaller size and phagocytic nature they are called micropoliceman of blood	Macropoliceman	
Diagram					



Plasma

- Forms liquid part of blood.
- Pale yellow in colour because of bilioubin.
- Constitute 55% of whole blood.
- In Plasma, there is 92 % water, 8% Organic & Inorganic Compounds.

Organic Compounds (Plasma Protein)

(i) **Albumin**— Maintain BCOP (28-32mm of Hg)

Blood colloidal osmotic pressure formed by liver.

(ii) **Globulin**— Acts as defence system in plasma
α, β, γ globulin
Fight with infections substance.

Globulin

- IgA**— Comes from mother's first milk cholestrom milk
- IgG**— Comes from placenta during pregnancy
- IgM**— First antibody appears after incident of any infection.

Blood			
Plasma (55%)	Blood Corpuscles (45%)		
	Red blood corpuscles	White blood corpuscles	Platelets



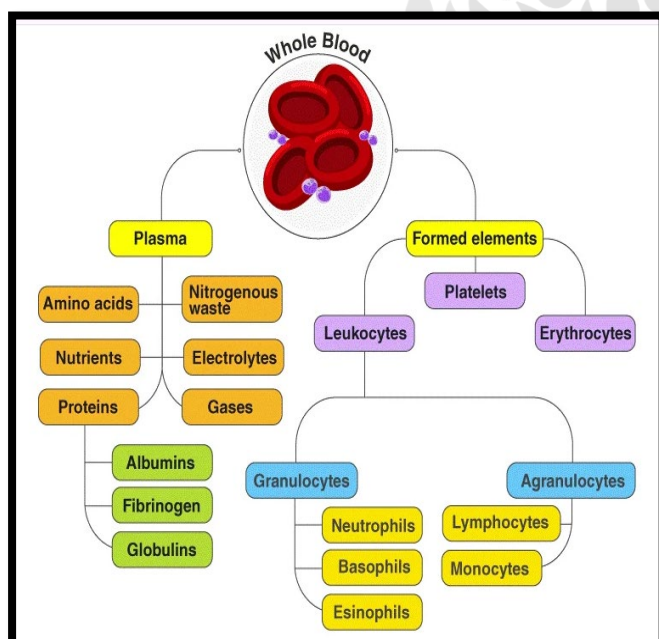
4. **IgE**—Fights with allergens.

(iii) **Prothrombin**

(iv) **Profibrinogen**

All types of plasma protein formed by liver.

- Plasma constitutes 55% of blood volume. It is a viscous fluid and contains 90- 92% water. Other than water, plasma contains 6-8% proteins (fibrinogens, albumins and globulins), amino acids, glucose and small amounts of electrolytes (Na^+ , Ca^{++} , Cl^- , etc.).
- Formed Elements include erythrocytes, leukocytes and blood platelets



Red blood corpuscles (RBC)

Red Blood Cells (RBCs) are the most abundant cells in human blood. They are also called erythrocytes.

Their main function is to transport oxygen from the lungs to body tissues and carry carbon dioxide back to the lungs.

Erythrocytes or RBCs are the most abundant cells present in the blood

- formed in the bone marrow
- Normal count $\rightarrow$ 5.5 million / mm^3 of blood
- biconcave and without a nucleus

have a lifespan of 120 days and get destroyed in the spleen (RBCs' graveyard)



RBC Contains millions of Respiratory pigments known as Hemoglobin.

Hemoglobin- Haem $\rightarrow$ 5-6% (contains Fe^{+3})

Globin : 95% $\rightarrow$ Protein (made up of α & β chain)

Function of Hemoglobin :

(i) Responsible for transportation of respiratory gases.

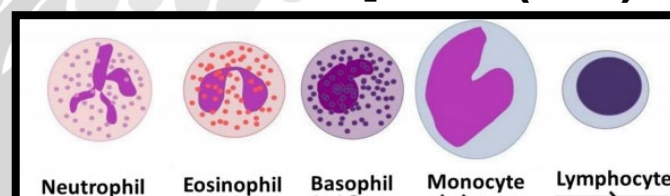
(a) Oxygen

(b) Carbon dioxide

$\text{Hb} + \text{O}_2 \rightleftharpoons \text{oxy hemoglobin}$ (unstable compound)

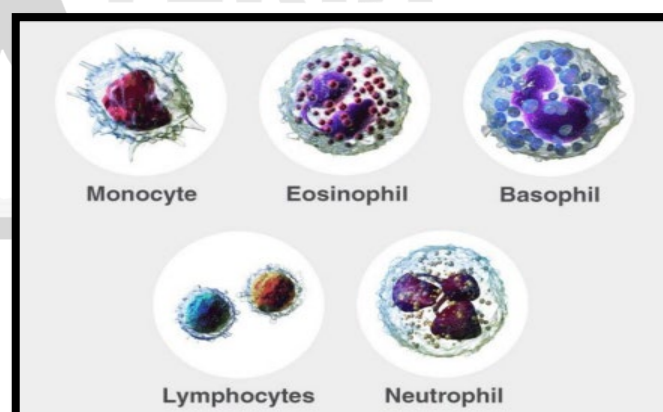
$\text{Hb} + \text{CO}_2 \rightleftharpoons \text{Carb amino hemoglobin}$

White blood Corpuscles (WBC)



Leucocytes or WBCs are colorless due to the absence of hemoglobin. They are of two types:

1. Granulocytes and
2. Agranulocytes.



- Have a life span of 3-4 days
- Granulocytes include neutrophils, eosinophils and basophils
- Agranulocytes include lymphocytes and monocytes



- Neutrophils have a polymorphic nucleus and have phagocytic ability
- Basophils are involved in inflammatory reactions as they secrete histamine, serotonin
- Eosinophils are involved in an allergic reaction

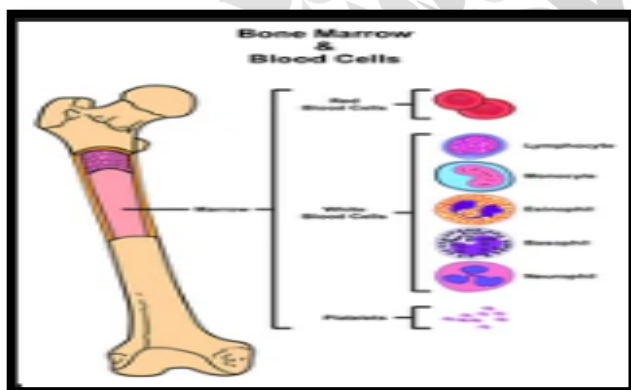
Blood Platelets

Platelets are also called thrombocytes. They are small, colorless cell fragments in blood that play a crucial role in blood clotting (coagulation) and prevention of bleeding.

Blood platelets or thrombocytes are involved in the clotting of blood. They are formed from the megakaryocyte cells of bone marrow.

General Characteristics of Platelets

- Shape: Irregular, disc-like fragments
- Nucleus: Absent
- Size: 2–4 μm
- Color: Colorless
- Origin: Formed from large cells called megakaryocytes



Mechanism of blood clotting

(Enzyme Cascade theory)

- Proposed by Macfarlane & Co-workers.
- According to this theory there are 3 steps in blood clotting.

1. Releasing of Thromboplastin:-

- ❖ Injured tissue synthesize exothromboplastin and platelets synthesize endothromboplastin.
- ❖ Both these thromboplastin react with plasma proteins in the presence of Ca^{2+} ions to form **Prothrombinase enzymes**. (Thrombokinase)
- ❖ This enzyme inactivate heparin. (Antiheparin)

2. Conversion of Prothrombin into Thrombin

- ❖ Prothrombinase enzyme convert inactive prothrombin into active thrombin in the presence of Ca^{2+} .

3. Conversion of fibrinogen into fibrin

- ❖ Fibrinogen is soluble protein of plasma. Thrombin protein polymerise monomers of fibrinogen to form insoluble fibrous protein fibrin.
- ❖ Fibrin fibres form network on cut or wound in which blood corpuscles got trapped. This form clotting of blood.
- ❖ After clotting a pale yellow liquid oozes from clot called Serum. In which antibodies are found.

Blood Corpuscles = Plasma
Plasma fibrinogen = Serum

Body Fluids and Circulation

Blood Groups:

BO grouping: There are four types of blood group, A, B, AB and O. It is based on the presence or absence of surface antigens on RBCs (antigen A and B).

ABO BLOOD GROUP SYSTEM				
	GROUP A	GROUP B	GROUP AB	GROUP O
RED BLOOD CELL TYPE				
ANTIBODIES IN PLASMA	Anti-B	Anti-A	None	Anti-A and Anti-B
ANTIGENS IN RED BLOOD CELL	A Antigen	B Antigen	A and B Antigens	None

Blood Group	A	B	AB	O
Antigen on RBCs	A	B	A,B	NIL
Antibodies in plasma	Anti-B	Anti-A	NIL	Anti-A Anti-B
Donor blood type	A,O	B,O	A, B, AB, O (universal acceptor)	O (universal donor)

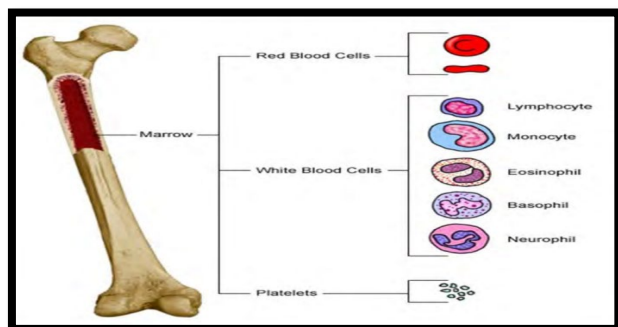
Rh grouping:

80% of people have Rh antigen on the surface of RBC. Those who have Rh antigen are called Rh⁺ and those without it are Rh⁻.





- Erythroblastosis fetalis: When the mother is Rh- and the father is Rh+ and the foetus blood group is Rh+, then there are chances of mixing of foetus' blood with that of the mother's blood during delivery.



- The mother's body produces antibodies against the antigen, which may go to the foetus and result in the agglutination of RBCs in the subsequent pregnancy.
- To avoid this mother is given anti-Rh antibodies after the delivery of the first child.

Coagulation of Blood

- Blood clotting prevents excess loss of blood from an injury. The clot is made up of a network of fibrins, which traps dead formed elements.
- Inactive fibrinogen → Fibrin by the enzyme Thrombin
- Inactive prothrombin → Thrombin by the enzyme Thrombokinase
- An injury stimulates platelets to release certain factors, which initiate the blood coagulation process.
- Ca⁺⁺ plays an important role in the process of coagulation.

Lymph

- Lymph is also a fluid connective tissue.
- The lymphatic system drains back the interstitial fluid back to major veins.
- Lymph consists of lymphocyte cells and is part of the immune system of the body.
- Fats are absorbed in the intestine by lacteals of villi and transported to the blood by lymph.

Antigen

- Blood Group Antigen are present on the surface of RBC
- These are chemical configurations which can produce immune response
- Also known as agglutinin.
- It triggers immune response in the body

Antibody

- Chemical configurations produce in response against antigen
- Are produced by lymphocyte WBC cell
- Also known as agglutinin.
- Produced by B-Lymphocyte

Difference between Antigen and Antibody

Antigen	Antibody
Foreign substance	Protein molecule
Triggers immune response	Fights antigen
Outside body	Produced inside body
Has epitopes	Has binding sites
Virus, bacteria	IgG, IgM

Blood Group System in Human

ABO Blood	Rh Blood
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Blood Transfusion

Blood group	Donate blood to	Received blood from
A	A and AB	A
B	B and AB	B
AB	AB	A,B, AB and O
O	A,B, AB and O	Only O

Genes of blood group

Blood Group	Genotype
A	I ^A I ^A or I ^A I ^O
B	I ^B I ^B or I ^B I ^O
AB	I ^A I ^B
O	I ^O I ^O

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